

VDI Migration

VMware Horizon to KVM + SPICE/RDP Desktop Virtualization

VMware Horizon VDI costs \$200-\$400/user/year. KVM with SPICE delivers the same experience at a fraction of the cost. This guide covers display protocol comparison (SPICE vs Blast vs RDP), GPU sharing for graphics desktops, connection broker options, and step-by-step migration from Horizon pools to KVM desktop VMs.

SPICE | RDP | NICE DCV | vGPU VDI | Connection Brokers | Thin Clients | \$150/user Savings

VDI Cost Comparison

VMware Horizon licensing is the most expensive component of VDI — and the easiest to eliminate.

Per-User VDI Cost (200 Users, 3 Years)

Component	VMware Horizon	KVM + SPICE	KVM + NICE DCV	Notes
VDI broker license	\$250/user/yr	\$0	\$0	Apache Guacamole or oVirt (free)
Hypervisor license	\$48/user/yr	\$0	\$0	KVM included in Linux kernel
Display protocol	Included (Blast)	\$0 (SPICE)	\$50/user/yr (DCV)	SPICE free; DCV for high-end graphics
GPU license (vGPU)	\$150/user/yr	\$150/user/yr	\$150/user/yr	Same NVIDIA vGPU cost either way
Windows license (VDA)	\$100/user/yr	\$100/user/yr	\$100/user/yr	Microsoft VDA required regardless
Server hardware (amortized)	\$50/user/yr	\$50/user/yr	\$50/user/yr	Same hardware
Per-User Annual	\$598	\$300	\$350	
200 Users, 3 Years	\$358,800	\$180,000	\$210,000	
3-Year Savings	—	\$178,800 (50%)	\$148,800 (41%)	

Without GPU (Office Workers): Even Cheaper

Remove vGPU for knowledge workers who don't need 3D graphics: VMware Horizon = \$348/user/yr vs KVM + SPICE = \$150/user/yr. **200 users, 3 years: save \$237,600 (57%).** SPICE handles office apps, web browsing, and video playback smoothly without GPU acceleration.

Display Protocol Comparison

SPICE, RDP, Blast, and NICE DCV — choosing the right protocol for your users.

Protocol	License	Video Quality	Latency	Bandwidth	USB Redirect	Multi-Monitor	Best For
SPICE	Free	Good	Good	Low-Med	Yes	Yes (4)	Office, web, general desktop
RDP	Free*	Good	Excellent	Very Low	Yes	Yes (16)	Windows desktops (built-in)
VMware Blast	\$250/user	Excellent	Excellent	Adaptive	Yes	Yes (4)	Horizon-only (vendor lock-in)
NICE DCV	\$50/user	Excellent	Excellent	Adaptive	Yes	Yes (4)	CAD, 3D rendering, GPU desktops
Apache Guacamole	Free	Good	Good	Med	No	No	Web-browser access (HTML5)
Parsec	\$30/user	Excellent	Excellent	Adaptive	Limited	Yes	Gaming, real-time graphics

* RDP included with Windows — no additional cost if using Windows VMs with RDP licenses.

Recommended Protocol by Use Case

Office Workers (No GPU)

- **Protocol:** RDP (Windows) or SPICE (Linux)
- **Cost:** \$0 additional
- **Quality:** Full desktop, multi-monitor, USB
- **Bandwidth:** 1-5 Mbps per user
- **Client:** Windows RDP client, virt-viewer, Remmina
- **Users:** 200 office workers, call center, data entry

CAD/Engineering (GPU Required)

- **Protocol:** NICE DCV or SPICE + vGPU
- **Cost:** \$50/user/yr (DCV) + \$150/user/yr (vGPU)
- **Quality:** Full GPU acceleration, 4K support
- **Bandwidth:** 10-30 Mbps per user
- **Client:** NICE DCV client (Windows/Mac/Linux)
- **Users:** Engineers, architects, designers

Connection Broker Options

Replacing VMware Horizon Connection Server with open-source alternatives.

Broker Comparison

Broker	License	Protocol	Auth	Pool Mgmt	Web Client	Notes
Apache Guacamole	Free	RDP, VNC, SSH	LDAP, TOTP, SSO	Manual	HTML5	Clientless, web-only access, great for BYOD
oVirt User Portal	Free	SPICE, VNC	LDAP, Kerberos	Pools	Yes	Full VDI features, VMware-like pool management
Cockpit	Free	VNC, SPICE	PAM/LDAP	Manual	Yes	Lightweight, per-host management
Ravada VDI	Free	SPICE, HTML5	LDAP, SSO	Pools	Yes	Purpose-built VDI broker for KVM
Flexiant	\$\$	Multiple	LDAP, SAML	Pools	Yes	Commercial alternative with support

Apache Guacamole Setup

```
# Install Guacamole (Docker – easiest)
docker run -d --name guacamole-db -e MYSQL_ROOT_PASSWORD=secret -e MYSQL_DATABASE=guacamole mysql:8
docker run -d --name guacd guacamole/guacd
docker run -d --name guacamole --link guacd:guacd --link guacamole-db:mysql \
  -e MYSQL_DATABASE=guacamole -e MYSQL_USER=guacamole -e MYSQL_PASSWORD=secret \
  -p 8080:8080 guacamole/guacamole

# Access: https://guacamole-host:8080/guacamole
# Add RDP connections to each desktop VM
# Users authenticate via LDAP → get assigned VM → RDP session in browser
# No client software needed – works on any device with a web browser
```

Migration Workflow

Step-by-step: Horizon desktop pool to KVM + SPICE/RDP desktops.

Step 1: Export Horizon Master Image

- In Horizon Admin: identify master/golden image VM
- Export as OVA: `govc export.ovf -vm /DC/vm/golden-win11`
- Or export VMDK from datastore directly
- Includes: OS, apps, profile settings, GPOs
- Note: linked clones won't export — use full clone or master

Step 2: Convert with h2kvmctl

```
h2kvmctl convert \  
  --source golden-win11.vmdk \  
  --format qcow2 --compress \  
  --virtio-drivers /usr/share/virtio-win/virtio-win.iso \  
  --emit-domain-xml  
  
# Creates: golden-win11.qcow2 (template)
```

Step 3: Create Desktop Pool

```
#!/bin/bash  
# create-vdi-pool.sh — create N desktop VMs  
TEMPLATE="golden-win11.qcow2"  
POOL_SIZE=50  
BRIDGE="br-vdi"  
  
for i in $(seq 1 $POOL_SIZE); do  
  NAME="desktop-$(printf '%03d' $i)"  
  # Linked clone (fast, space-efficient)  
  qemu-img create -f qcow2 \  
    -b $TEMPLATE -F qcow2 \  
    /var/lib/libvirt/images/${NAME}.qcow2  
  # Define VM  
  virt-install --name $NAME \  
    --ram 4096 --vcpus 2 \  
    --import --disk /var/lib/libvirt/images/${NAME}.qcow2 \  
    --network bridge=$BRIDGE,model=virtio \  
    --graphics spice --noautoconsole  
done  
echo "Created $POOL_SIZE desktop VMs"
```

Step 4: Connect Users

- **SPICE:** `remote-viewer spice://kvm-host:5900`
- **RDP:** `mstsc /v:desktop-001.corp.local`
- **Guacamole:** Browser → login → click desktop name
- **Thin clients:** Configurable SPICE/RDP endpoints
- LDAP auth maps user → assigned desktop VM
- User profile stored on network share (roaming)

VMware Horizon's value is the connection broker + display protocol — both replaceable with free alternatives. SPICE provides excellent desktop experience for office users. RDP is built into Windows and works everywhere. Apache Guacamole adds web-browser access without any client install. For GPU desktops, NICE DCV at \$50/user beats Horizon at \$250/user. hyper2kvm converts golden images and linked clones from Horizon to KVM in minutes.